

**INSTRUMENT
APPROACH
CHART**

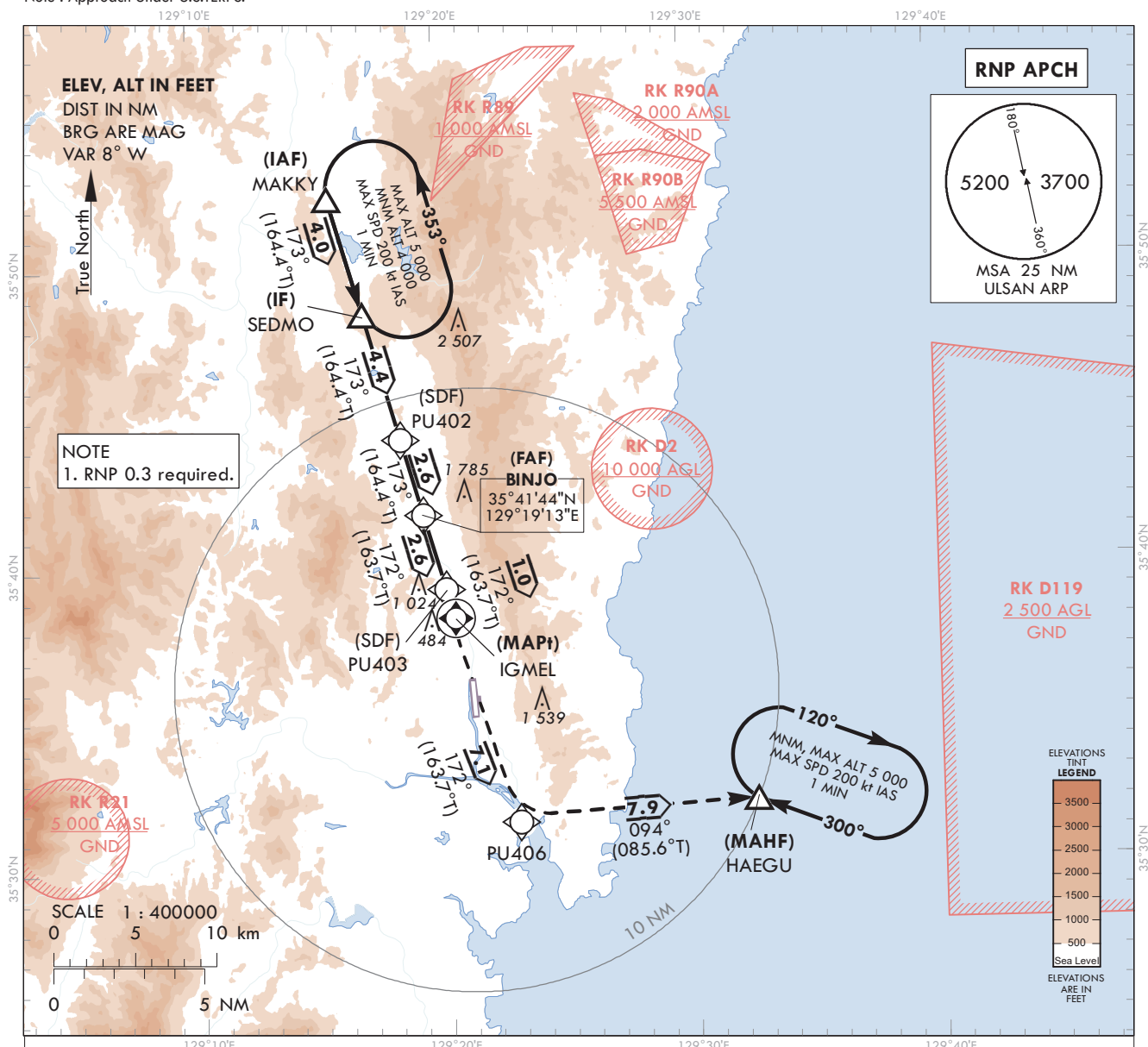
AERODROME ELEV 42 ft
HEIGHTS RELATED TO
THR RWY 18 - ELEV 42 ft

POHANG	APP	124.25
		120.2
ULSAN	TWR	118.75

ULSAN/Ulsan(RKP)

RNP Y RWY 18

Note : Approach Under U.S.TERPS.

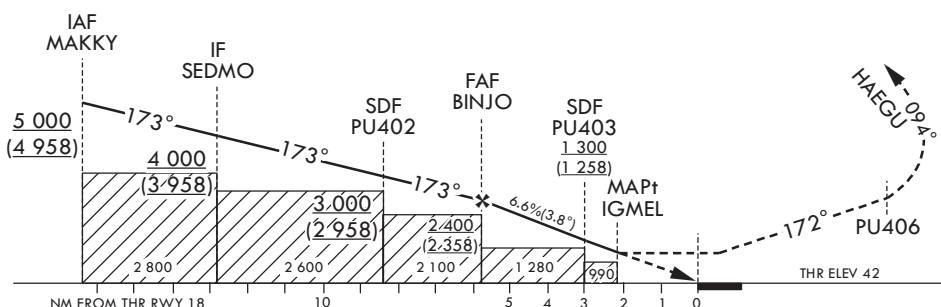


RECOMMENDED PROFILE(FAA)	THR RWY 18	5	4	3	2.2
Final Approach Gradient 6.58%, 400 ft/NM	ALT(HGT)	2 092 (2 050)	1 692 (1 650)	1 292 (1 250)	990 (948)

TRANSITION ALT 14 000
TRANSITION LVL FL 140

MISSED APPROACH

Climb on track of 172° to PU406 then LEFT turn on track of 094° to HAEGU and HOLD at 5 000 ft.



* Missed APCH climb rate :
304 ft/NM(5.0%)

OCA (H)		A	B	C	D	Rate of descent	V/V fpm	400	600	800	1 000	1 200
Straight-in Approach	LNAV	990 (948)			N/A	* Timing Not authorized for defining MAPt. * Circling Not authorized.						

Change : Information of airspace restrictions(RK D2 and D119).

AERONAUTICAL DATA TABULATION

Instrument Approach Procedure Coding Tables

RNP Y RWY 18

Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(T)	Distance (NM)	Turn direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks
001	IF	MAKKY	-	-	-	-	+5 000	-	35°52'18.8"N 129°15'35.8"E	-	RNP APCH	IAF
002	TF	SEDMO	-	173(164.4)	4.0	-	+4 000	-	35°48'27.1"N 129°16'55.0"E	-	RNP APCH	IF
003	TF	PU402	-	173(164.4)	4.4	-	+3 000	-	35°44'14.2"N 129°18'21.4"E	-	RNP APCH	SDF
004	TF	BINJO	-	173(164.4)	2.6	-	+2 400	-	35°41'43.5"N 129°19'12.8"E	-	RNP APCH	FAF
005	TF	PU403	-	172(163.7)	2.6	-	+1 300	-	35°39'15.5"N 129°20'05.6"E	-	RNP APCH	SDF
006	TF	IGMEL	Y	172(163.7)	1.0	-	+990	-	35°38'17.7"N 129°20'26.4"E	-3.77/50	RNP APCH	MAPt
007	TF	PU406	-	172(163.7)	7.1	-	-	-	35°31'29.1"N 129°22'52.3"E	-	RNP APCH	-
008	TF	HAEGU	Y	094(085.6)	7.9	-	-	-	35°32'06.7"N 129°32'32.3"E	-	RNP APCH	-
009	HIM	HAEGU	Y	300 (292.1)	-	R	@5 000	-200	35°32'06.7"N 129°32'32.3"E	-	RNP APCH	1 min (Outbound timing)

HOLDING PROCEDURE

Holding Identification	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(T)	Time (min)	Turn direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks
RNP Y RWY 18	HIM	SEDMO	Y	173(164.4)	1.0	L	-5 000 +4 000	-200	35°48'27.1"N 129°16'55.0"E	-	RNAV 1	-

**INSTRUMENT
APPROACH
CHART - ICAO**

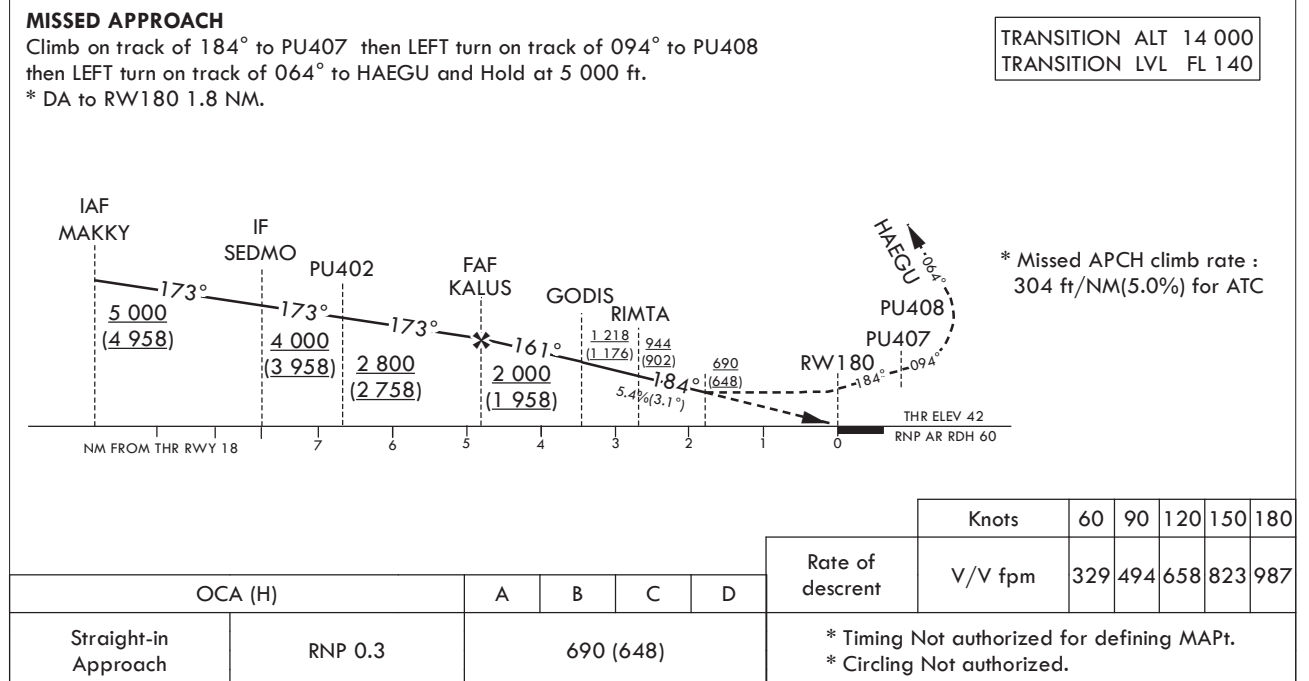
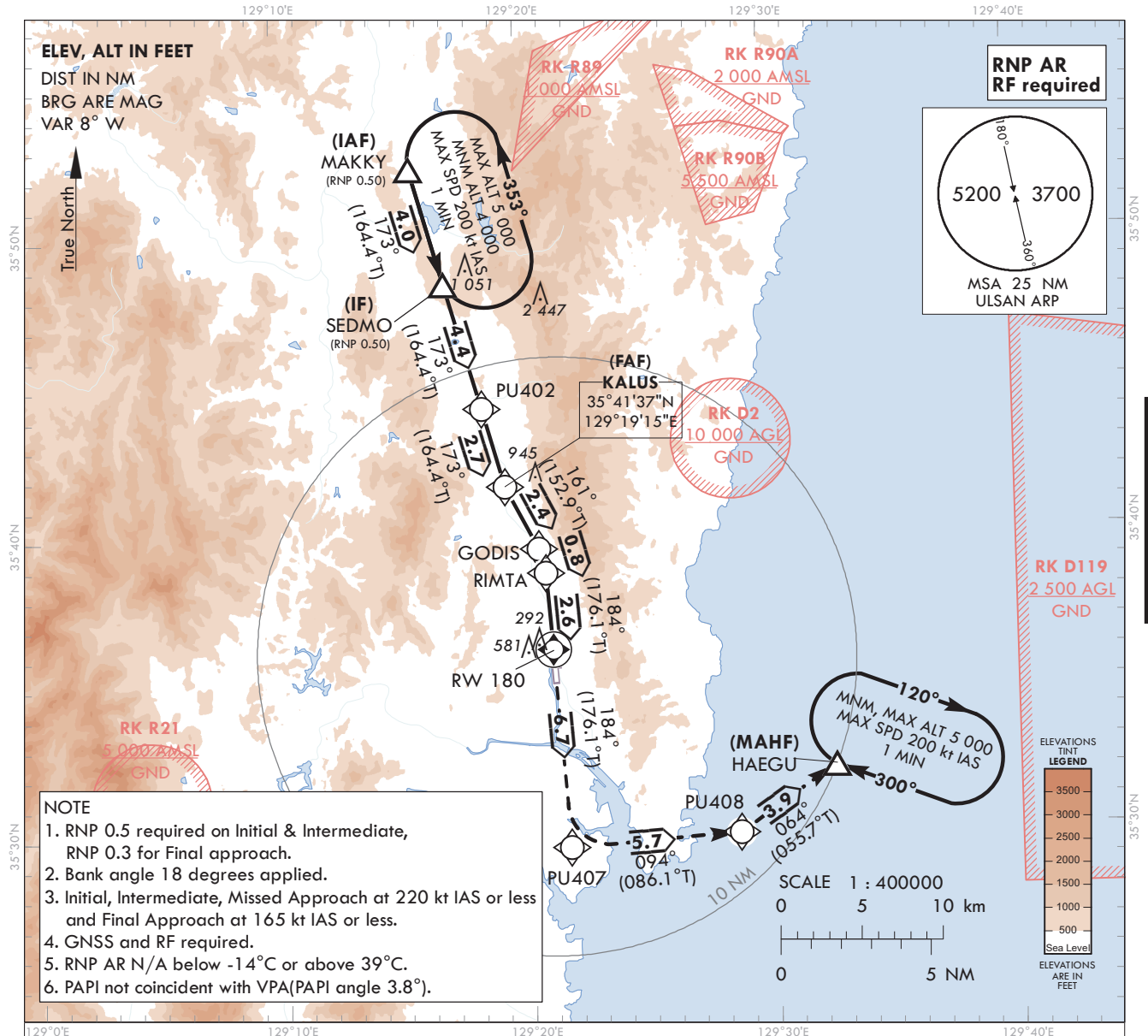
AERODROME ELEV 42 ft
HEIGHTS RELATED TO
THR RWY 18 - ELEV 42 ft

POHANG	APP	124.25
		120.2
ULSAN	TWR	118.75

ULSAN/Ulsan(RKPU)

RNP Z RWY 18(AR)

Note : Approach Under ICAO Flight Procedures.



Change : Information of airspace restrictions(RK D2 and D119).

AERONAUTICAL DATA TABULATION

Instrument Approach Procedure Coding Tables

RNP Z RWY 18 (AR)

Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(T)	Distance (NM)	Turn direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks
001	IF	MAKKY	-	-	-	-	+5 000	-220	35°52'18.8"N 129°15'35.8"E	-	-	IAF
002	TF	SEDWO	-	173(164.4)	4.0	-	+4 000	-220	35°48'27.1"N 129°16'55.0"E	-	0.5	IF
003	TF	PU402	-	173(164.4)	4.4	-	+2 800	-220	35°44'14.2"N 129°18'21.4"E	-	0.5	
004	TF	KALUS	-	173(164.4)	2.7	-	+2 000	-165	35°41'37.4"N 129°19'14.9"E	-	0.3	FAF Bank Angle 18°
005	TF	GODIS	-	161(152.9)	2.4	L	+1 218	-165	35°39'30.7"N 129°20'34.3"E	-3.10	0.3	Bank Angle 18°
006	RF Center : ARCO1 r=2.08 NM	RIMTA	-	-	0.8	R	+944	-165	35°38'42.4"N 129°20'50.7"E	-3.10	0.3	Bank Angle 18°
007	TF	RW180	Y	184(176.1)	2.6	-	-	-165	35°36'08.9"N 129°21'03.5"E	-3.10/60	0.3	-
008	TF	PU407	-	184(176.1)	6.7	-	-	-220	35°29'30.2"N 129°21'36.8"E	-	1.0	
009	TF	PU408	-	094(086.1)	5.7	L	-	-220	35°29'53.2"N 129°28'33.2"E	-	1.0	Bank Angle 18°
010	TF	HAEGU	Y	064(055.7)	3.9	L	+5 000	-220	35°32'06.7"N 129°32'32.3"E	-	1.0	Bank Angle 18°
011	HM	HAEGU	Y	300 (292.1)	-	R	@5 000	-200	35°32'06.7"N 129°32'32.3"E	-	RNP APCH	1 min (Outbound timing)

RF arc center Identifier	Coordinates
ARCO1	35°38'33.8"N 129°18'18.0"E

**INSTRUMENT
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CHART**

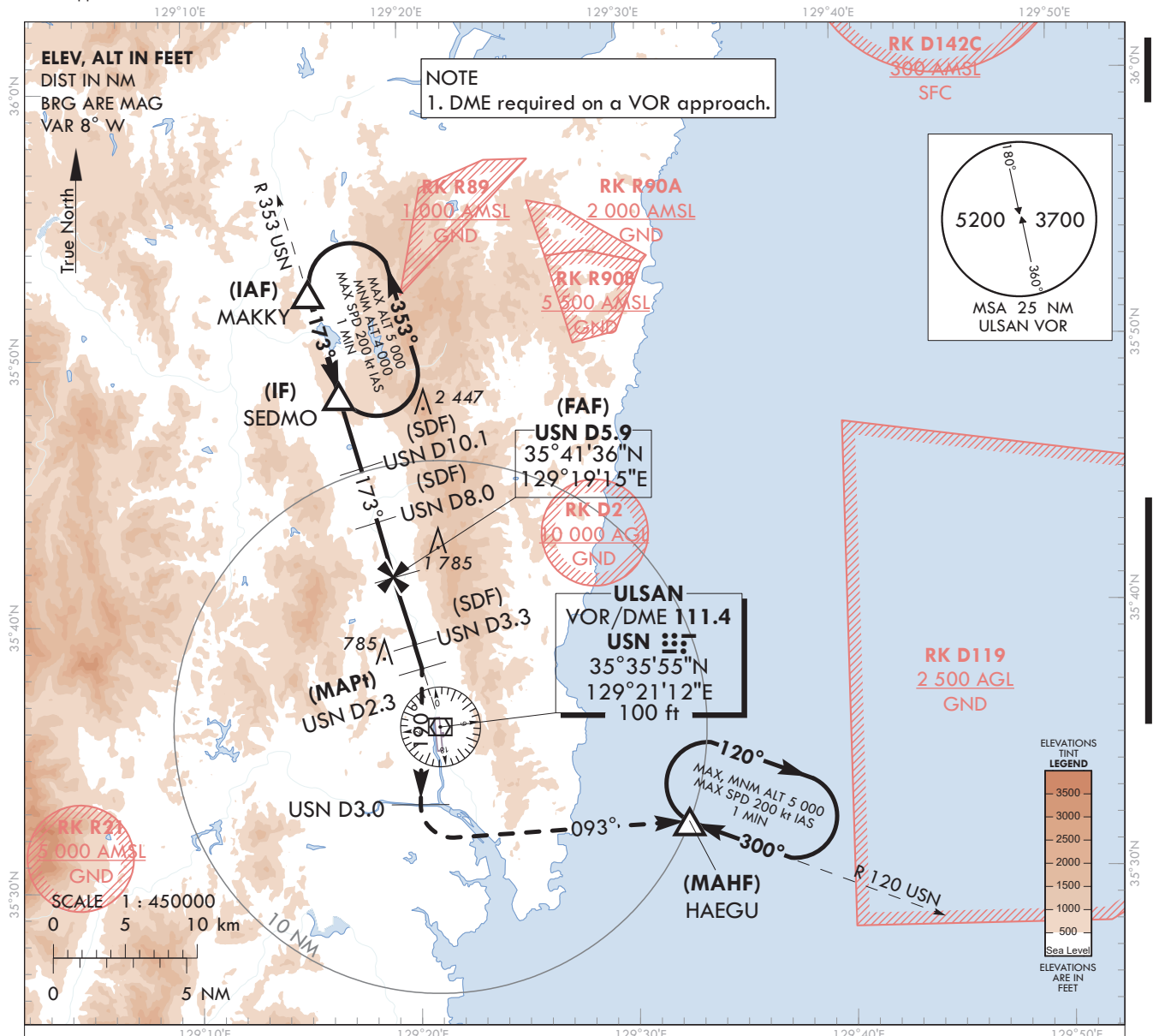
AERODROME ELEV 42 ft
HEIGHTS RELATED TO AD ELEV

POHANG APP 124.25
120.2
ULSAN TWR 118.75

ULSAN/Ulsan(RKPU)

VOR RWY 18

Note : Approach Under U.S. TERPS.



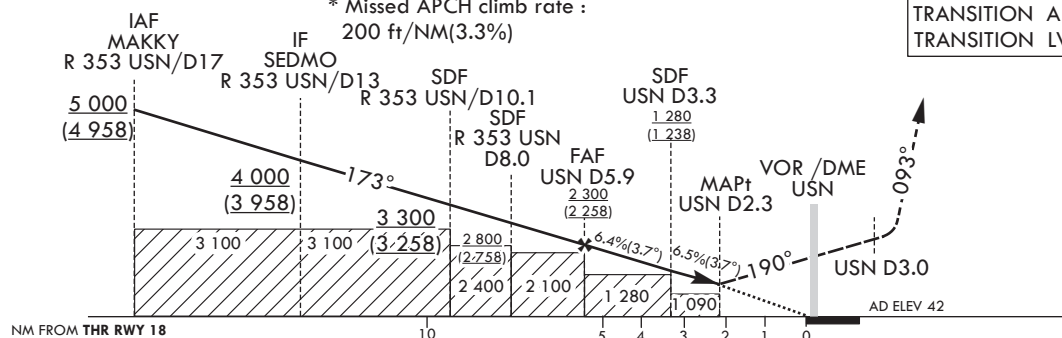
MISSED APPROACH

Climb on HDG 190° to USN 3 DME,
then LEFT turn direct to HAEGU
and Hold at 5 000 ft.

RECOMMENDED PROFILE (FAA)	DME USN	5	4	3	2.8
Final Approach Gradient 6.45%, 392 ft/NM	ALT (HGT)	1 947 (1 906)	1 556 (1 514)	1 164 (1 122)	1 080 (1 038)

* Missed APCH climb rate :
200 ft/NM (3.3%)

TRANSITION ALT 14 000
TRANSITION LVL FL 140



OCA (H)	A	B	C	D		Knots	60	90	120	150	180
Straight-in Approach	1 090 (1 048)			N/A	Rate of descent	V/V fpm	392	588	784	980	1 175
					* Timing Not authorized for defining MAPt. * Circling Not authorized.						

Change : Information of airspace restrictions (RK D2, D119 and D142C).

ULSAN/Ulsan(RKPU)
VOR RWY 18
AERONAUTICAL DATA TABULATION

VOR Approach to RWY 18 from MAKKY		
Fix/point		Coordinates
MAKKY(IAF)	R 353 USN / 17.00 NM USN	35°52'18.8"N 129°15'35.8"E
SEDMO(IF)	BRG 172.64° / 13.00 NM USN	35°48'27.1"N 129°16'55.0"E
D10.1 USN(SDF)	BRG 172.64° / 10.10 NM USN	35°45'39.1"N 129°17'51.8"E
D8.0 USN(SDF)	BRG 172.64° / 8.00 NM USN	35°43'37.6"N 129°18'33.5"E
D5.9 USN(FAF)	BRG 172.64° / 5.90 NM USN	35°41'36.0"N 129°19'15.1"E
D3.3 USN(SDF)	BRG 172.64° / 3.28 NM USN	35°39'04.4"N 129°20'07.0"E
D2.3 USN(MAPt)	BRG 172.64° / 2.26 NM USN	35°38'05.4"N 129°20'27.1"E
THR RWY 18	Final approach descent angle 3.67°(FAF-SDF) / 3.71°(SDF-THR)	35°36'08.89"N 129°21'03.54"E
USN VOR/DME		35°35'54.6"N 129°21'11.8"E
D3.0 USN	BRG 190.62° / 3.00 NM USN	35°33'01.3"N 129°20'11.6"E
HAEGU	BRG 93.43° / 10.00 NM USN	35°32'06.7"N 129°32'32.3"E

**INSTRUMENT
APPROACH
CHART - ICAO**

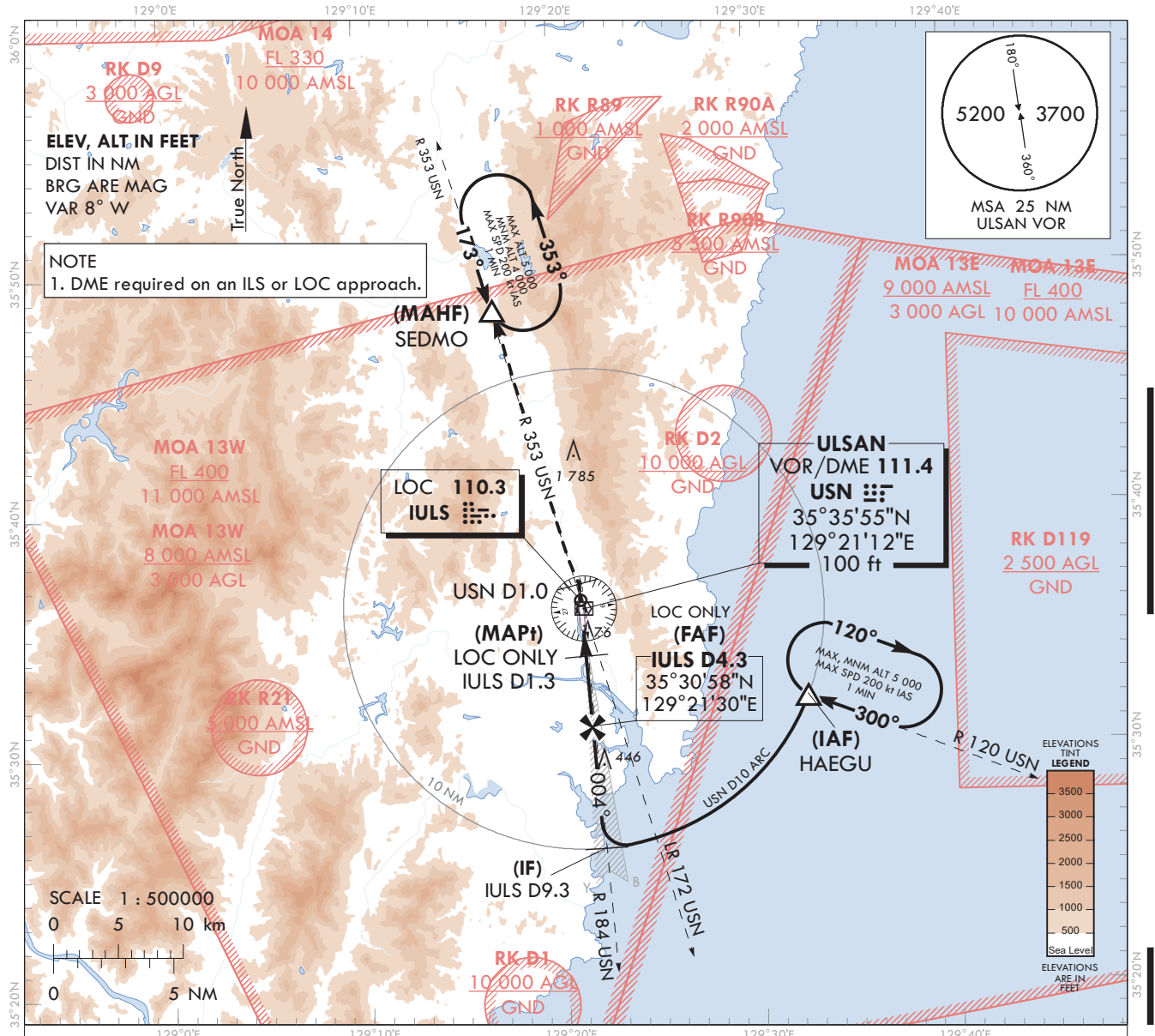
AERODROME ELEV 42 ft
HEIGHTS RELATED TO
THR RWY 36 - ELEV 28 ft

POHANG APP 124.25
120.2
ULSAN TWR 118.75

ULSAN/Ulsan(RKPU)

ILS Y or LOC Y RWY 36

Note : Approach under ICAO Flight Procedures.



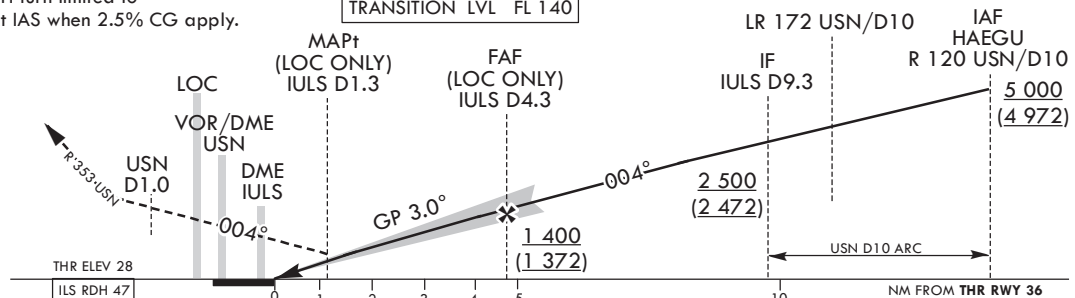
MISSED APPROACH

Climb on HDG 004° to USN 1 DME,
then LEFT turn to intercept R 353 USN to SEDMO.
Hold at SEDMO at 4 000 ft.

* Missed APCH turn limited to
MAX 200 kt IAS when 2.5% CG apply.

RECOMMENDED PROFILE(LOC ONLY)	DME IULS	6	5	4	3	2.3
Final Approach Gradient 5.28%, 321 ft/NM	ALT(HGT)	1 950 (1 922)	1 620 (1 592)	1 300 (1 272)	980 (952)	750 (722)

TRANSITION ALT 14 000
TRANSITION LVL FL 140



OCA (H)		A	B	C	D							
Straight-in Approach	CAT-I (CG 2.5%)	710 (682)				Rate of descent	Knots	60	90	120	150	180
	CAT-I (CG 5.0%)	246 (218)					V/V fpm	318	478	637	796	955
	LOC	750 (722)					*Timing Not authorized for defining MAPt. *Circling Not authorized.					

Change : Information of airspace restrictions(RK D1, D2 and D119).

AERONAUTICAL DATA TABULATION

ILS Y / LOC Y Approach to RWY 36 from HAEGU			
Fix/point		Coordinates	
HAEGU(IAF)	R 120 USN / 10.00 NM USN	35°32'06.7"N	129°32'32.3"E
D9.3 IULS(IF)	BRG 4.31° / 9.30 NM IULS	35°25'54.7"N	129°21'54.8"E
D4.3 IULS(FAF LOC ONLY)	BRG 4.31° / 4.30 NM IULS	35°30'57.6"N	129°21'29.5"E
D1.3 IULS(MAPt LOC ONLY)	BRG 4.31° / 1.30 NM IULS	35°33'55.7"N	129°21'14.7"E
THR RWY 36	Final approach descent angle 2.90° (LOC ONLY)	35°35'04.15"N	129°21'08.95"E
IULS DME		35°35'13.8"N	129°21'12.2"E
USN VOR/DME		35°35'54.6"N	129°21'11.8"E
IULS LOC		35°36'16.3"N	129°21'02.9"E
D1.0 USN	BRG 4.31° / 1.00 NM USN	35°36'53.8"N	129°20'59.8"E
SEDMO	BRG 352.68° / 13.00 NM USN	35°48'27.1"N	129°16'55.0"E

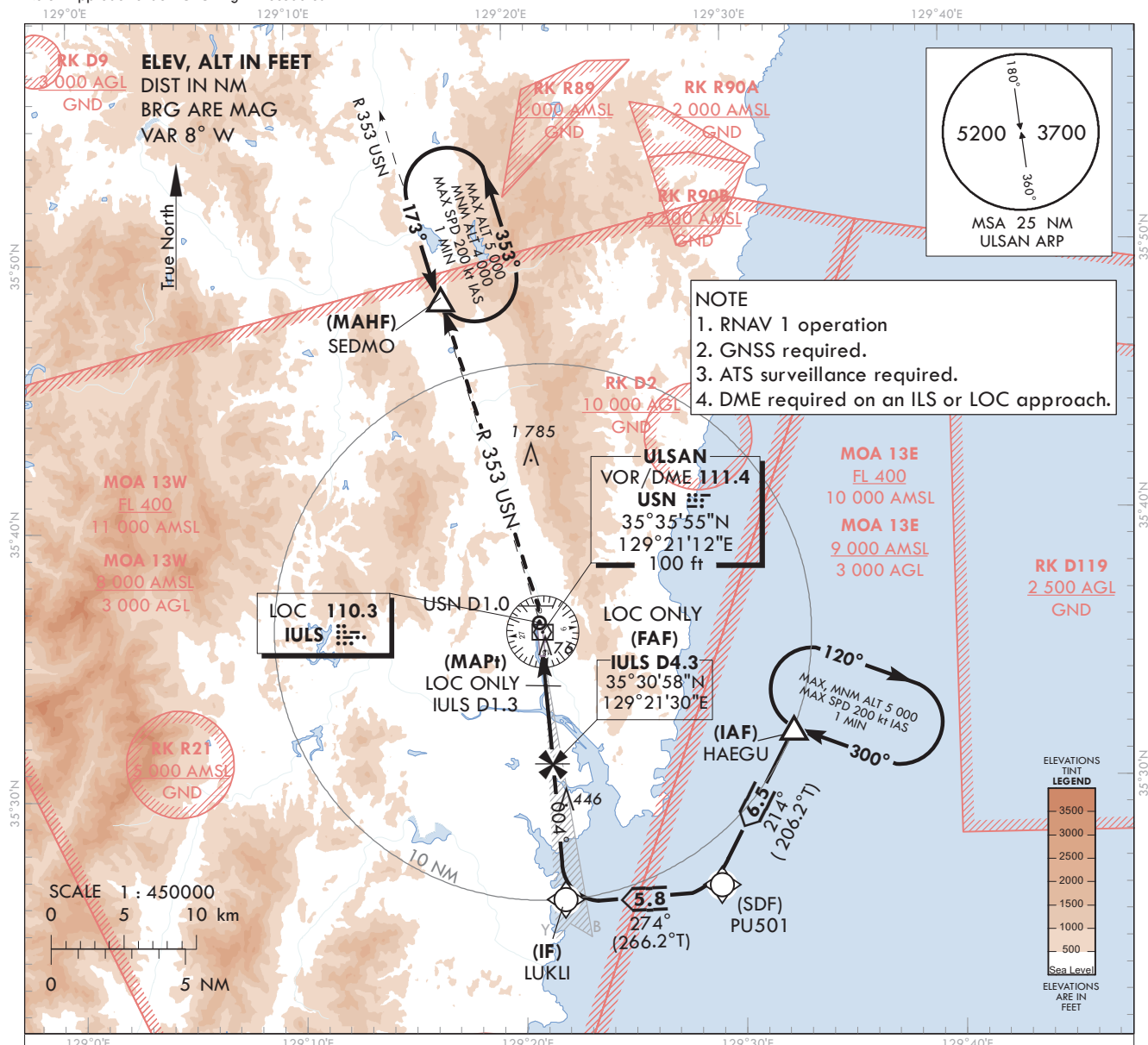
AERODROME ELEV 42 ft
HEIGHTS RELATED TO
THR RWY 36 - ELEV **28 ft**

POHANG	APP	124.25
		120.2
ULSAN	TWR	118.75

ULSAN/Ulsan(RKPU)

ILS Z or LOC Z RWY 36

Note : Approach under ICAO Flight Procedures.

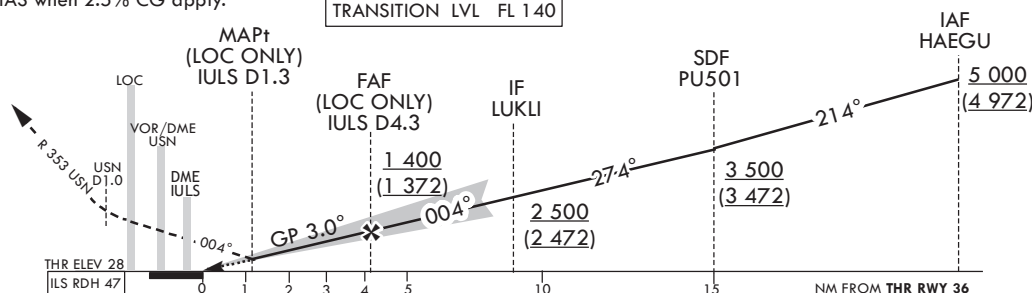


Climb on HDG 004° to USN 1 DME, then LEFT turn to intercept R 353 USN to SEDMO.
Hold at SEDMO at 4 000 ft.

* Missed APCH turn limited to
MAX 200 kt IAS when 2.5% CG apply.

RECOMMENDED PROFILE(LOC ONLY)	DME IULS	6	5	4	3	2.3
Final Approach Gradient 5.28%, 321 ft/NM	ALT(HGT)	1 950 (1 922)	1 620 (1 592)	1 300 (1 272)	980 (952)	750 (722)

TRANSITION	ALT	14 000
TRANSITION	LVL	FL 140



OCA (H)		A	B	C	D							
Straight-in Approach	CAT-I (CG 2.5%)	710 (682)				Rate of descent	Knots	60	90	120	150	180
	CAT-I (CG 5.0%)	246 (218)					V/V fpm	318	478	637	796	955
	LOC	750 (722)				*Timing Not authorized for defining MAPt. *Circling Not authorized.						

Change : Information of airspace restrictions(RK D2 and D119).

AERONAUTICAL DATA TABULATION

ILS Z / LOC Z Approach to RWY 36 from HAEGU			
Fix/point		Coordinates	
HAEGU(IAF)		35°32'06.7"N	129°32'32.3"E
PU501(SDF)		35°26'18.1"N	129°29'02.9"E
LUKLI(IF)	BRG 4.31° / 9.30 NM IULS	35°25'54.4"N	129°21'54.8"E
D4.3 IULS(FAF LOC ONLY)	BRG 4.31° / 4.30 NM IULS	35°30'57.6"N	129°21'29.5"E
D1.3 IULS(MAPt LOC ONLY)	BRG 4.31° / 1.30 NM IULS	35°33'55.7"N	129°21'14.7"E
THR RWY 36	Final approach descent angle 2.90° (LOC ONLY)	35°35'04.15"N	129°21'08.95"E
IULS DME		35°35'13.8"N	129°21'12.2"E
USN VOR/DME		35°35'54.6"N	129°21'11.8"E
IULS LOC		35°36'16.3"N	129°21'02.9"E
D1.0 USN	BRG 4.31° / 1.00 NM USN	35°36'53.8"N	129°20'59.8"E
SEDMO	BRG 352.68° / 13.00 NM USN	35°48'27.1"N	129°16'55.0"E

**INSTRUMENT
APPROACH
CHART - ICAO**

AERODROME ELEV 42 ft
HEIGHTS RELATED TO
THR RWY 36 - ELEV 28 ft

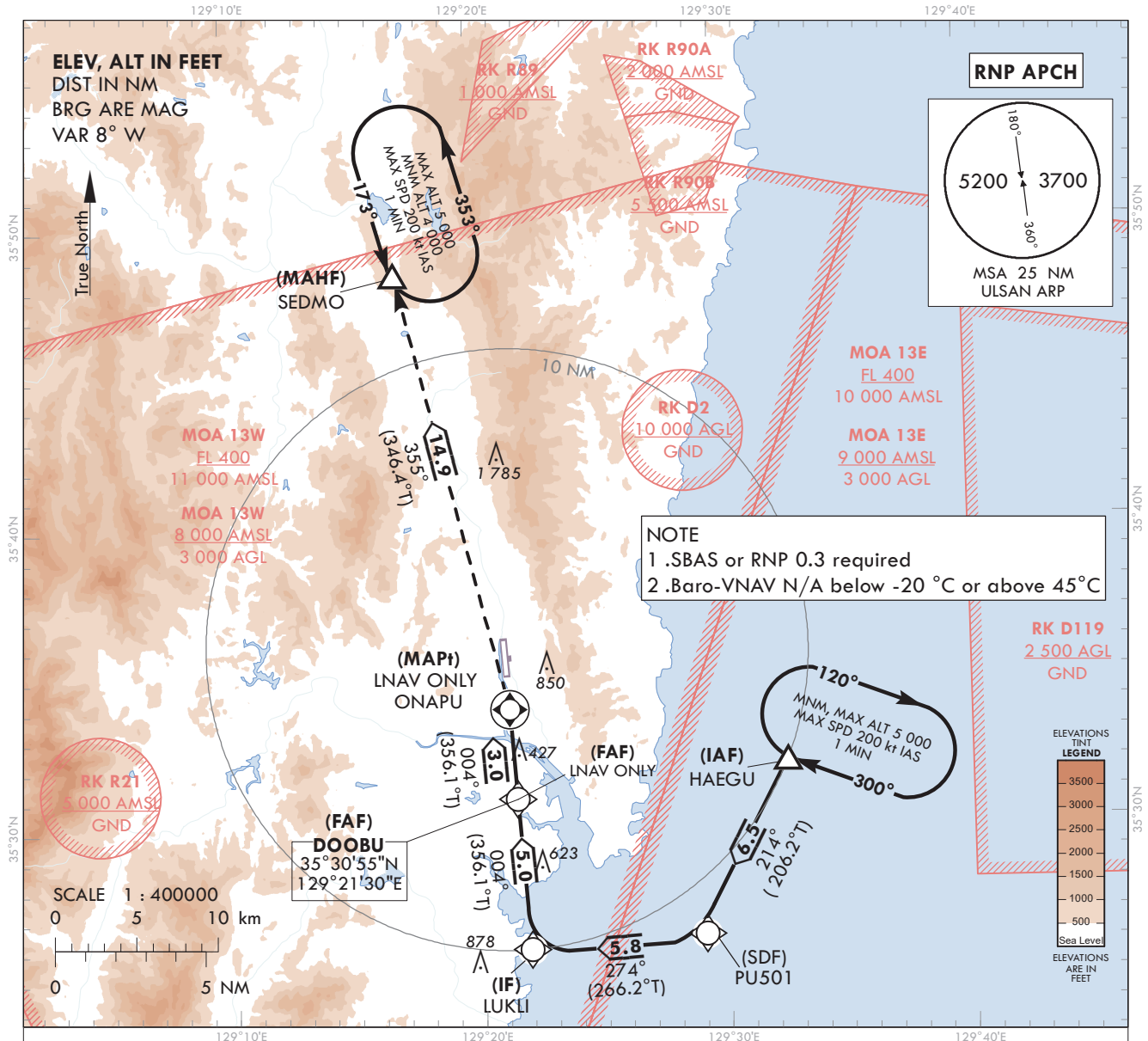
POHANG APP 124.25
ULSAN TWR 118.75

KASS
CH1318
K36A
RDH: 50

ULSAN/Ulsan(RKPU)

RNP RWY 36

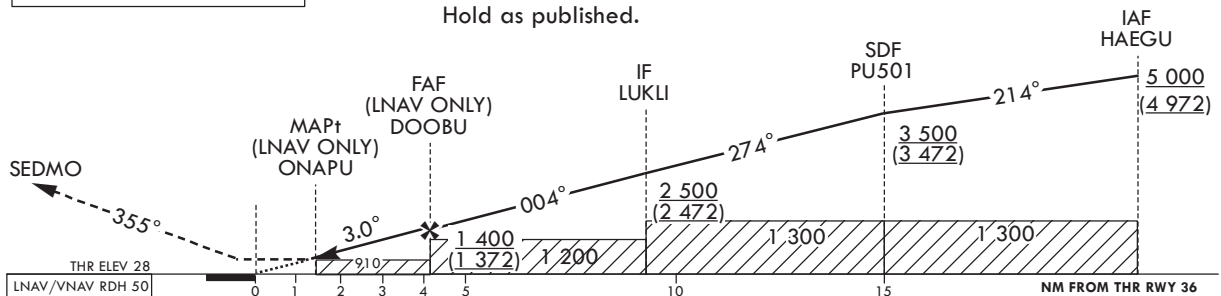
Note : Approach under ICAO Flight Procedures.



TRANSITION ALT 14 000
TRANSITION LVL FL 140

MISSED APPROACH

Climb to 4 000 ft on track of 355° to SEDMO.
Hold as published.



OCA (H)		A	B	C	D						
Straight-in Approach	LPV (CG 2.5%)	740 (712)			Rate of descent	Knots	60	90	120	150	180
	LPV (CG 5.0%)	340 (312)				V/V fpm	319	478	637	796	956
	LNAV/VNAV	810 (782)				* Timing Not authorized for defining MAPt. * Circling Not authorized.					
	LNAV	910 (882)									

Change : Information of airspace restrictions(RK D2 and D119).

AERONAUTICAL DATA TABULATION

Instrument Approach Procedure Coding Tables

RNP RWY 36

Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(T)	Distance (NM)	Turn direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/ RDH	Navigation specification	Remarks
001	IF	HAEGU	-	-	-	-	+5 000	-	35°32'06.7"N 129°32'32.3"E	-	RNP APCH	IAF
002	TF	PU501	-	214(206.2)	6.5	-	+3 500	-	35°26'18.1"N 129°29'02.9"E	-	RNP APCH	SDF
003	TF	LUKLI	-	274(266.2)	5.8	-	+2 500	-	35°25'54.4"N 129°21'54.8"E	-	RNP APCH	IF
004	TF	DOOBU	-	004(356.1)	5.0	-	+1 400	-	35°30'55.3"N 129°21'29.7"E	-	RNP APCH	FAF
005	TF	ONAPU	Y	004(356.1)	3.0	-	+910	-	35°33'54.4"N 129°21'14.8"E	-3.00/50	RNP APCH	MAPt
006	HM	SEDMO	Y	355(346.4)	14.9	-	-	-	35°48'27.1"N 129°16'55.0"E	-	RNP APCH	1 min (Outbound timing)

HOLDING PROCEDURE

Holding Identification	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(T)	Time (min)	Turn direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/ RDH	Navigation specification	Remarks
RNP RWY 36	HM	HAEGU	Y	300 (292.1)	1.0	R	@5 000	-200	35°32'06.7"N 129°32'32.3"E	-	RNAV 1	-
	HM	SEDMO	Y	173 (164.4)	1.0	L	-5 000 +4 000	-200	35°48'27.1"N 129°16'55.0"E	-	RNAV 1	-

ULSAN/Ulsan(RKPU)
RNP RWY 36

FAS DATA BLOCK INFORMATION

INPUT DATA

Parameters	Values
Operation Type	0
SBAS Provider	6 (KASS)
Airport Identifier	RKPU
Runway	36
Runway Letter	0 (None)
Approach Performance Designator	0
Route Indicator	
Reference Path Data Selector	0
Reference Path Identifier	K36A
LTP/FTP Latitude	353504.1470N
LTP/FTP Longitude	1292108.9500E
LTP/FTP Ellipsoidal Height (meters)	+00383(+38.3m)
FPAP Latitude	353608.8905N
Delta FPAP Latitude (seconds)	64.7435
FPAP Longitude	1292103.5405E
Delta FPAP Longitude (seconds)	-5.4095
Threshold Crossing Height (TCH)	50.0(feet) / 15.2(m)
TCH Units Selector (Feet)	0
Glidepath Angle (GPA, degrees)	3.00
Course Width (metres)	105.00
Length Offset (metres)	0
Horizontal Alert Limit (HAL, meters)	40.0
Vertical Alert Limit (VAL, meters)	50.0

OUTPUT DATA

Data Block	60 15 10 0B 12 24 00 00 01 36 33 0B A6 6D 45 0F AC 16 83 37 7F 15 CF F9 01 BD D5 FF F4 01 2C 01 64 00 C8 FA 10 38 07 E6
Calculated CRC Value	103807E6

INTENTIONALLY

LEFT

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**INSTRUMENT
APPROACH
CHART - ICAO**

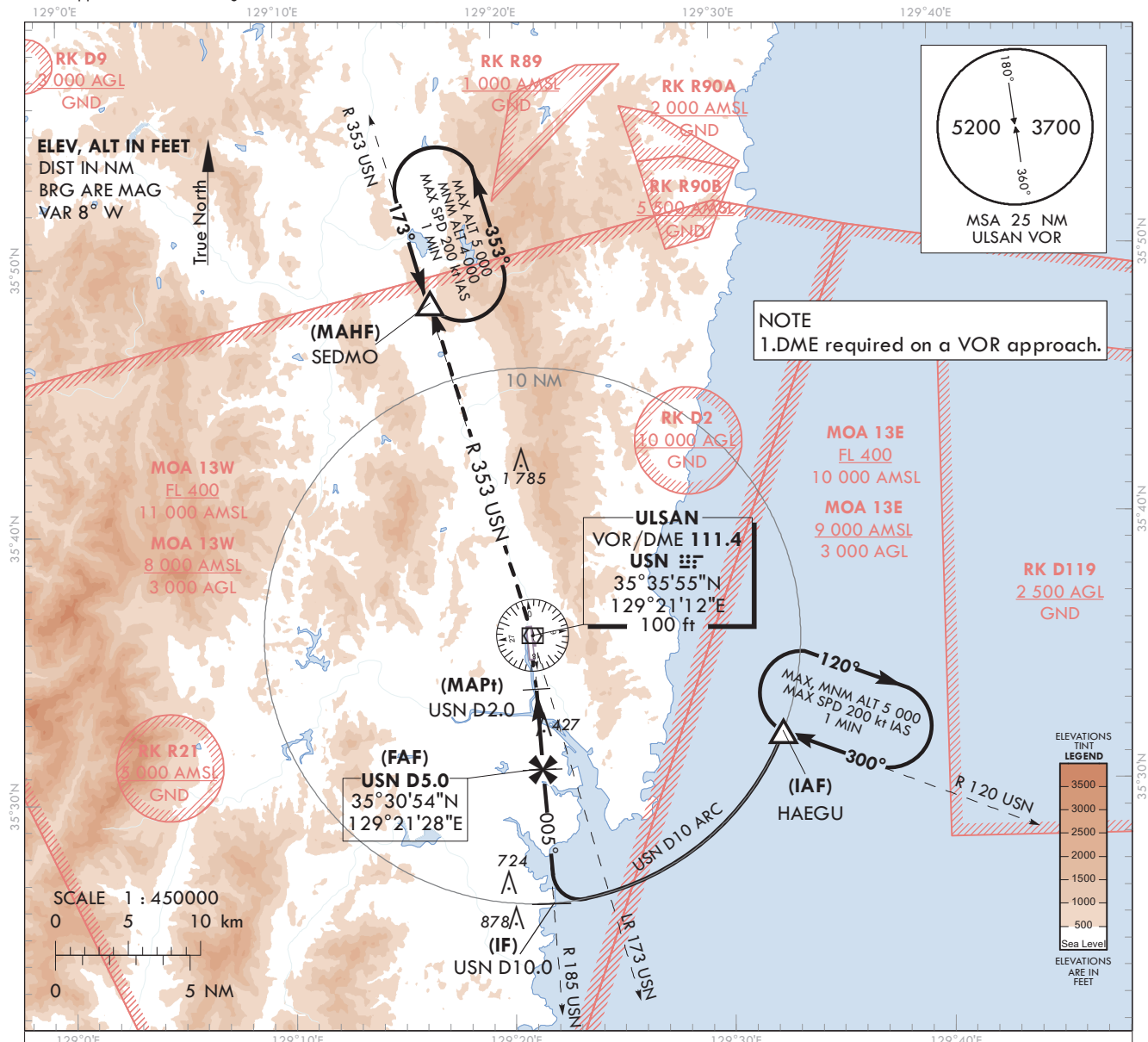
AERODROME ELEV 42 ft
HEIGHTS RELATED TO
THR RWY 36 - ELEV 28 ft

POHANG	APP	124.25
		120.2
ULSAN	TWR	118.75

ULSAN/Ulsan(RKPU)

VOR RWY 36

Note : Approach under ICAO Flight Procedures.

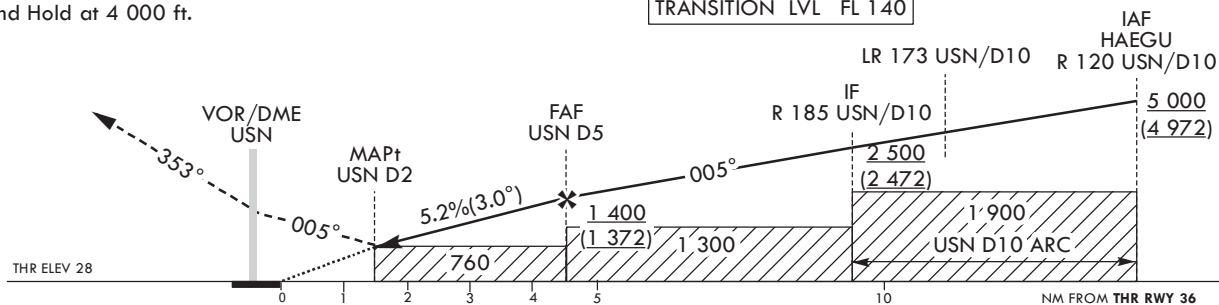


RECOMMENDED PROFILE	DME USN	5	4	3
Final Approach Gradient 5.24%, 318 ft/NM	ALT(HGT)	1 400 (1 372)	1 080 (1 052)	760 (732)

MISSED APPROACH

Climb direct to USN then proceed on R 353 USN to SEDMO and Hold at 4 000 ft.

TRANSITION ALT 14 000
TRANSITION LVL FL 140



OCA (H)	A	B	C	D		Knots	60	90	120	150	180
Straight-in Approach	760 (732)				Rate of descent	V/V fpm	318	477	636	795	954
					* Timing Not authorized for defining MAPt. * Circling Not authorized.						

Change : Information of airspace restrictions(RK D2 and D119).

ULSAN/Ulsan(RKPU)
VOR RWY 36
AERONAUTICAL DATA TABULATION

VOR Approach to RWY 36 from HAEGU		
Fix/point		Coordinates
HAEGU(IAF)	R 120 USN / 10.00 NM USN	35°32'06.7"N 129°32'32.3"E
D10.0 USN(IF)	BRG 5.63° / 10.00 NM USN	35°25'54.3"N 129°21'44.9"E
D5.0 USN(FAF)	BRG 5.63° / 5.00 NM USN	35°30'54.4"N 129°21'28.4"E
D2.0 USN(MAPt)	BRG 5.63° / 2.00 NM USN	35°33'54.5"N 129°21'18.5"E
THR RWY 36	Final approach descent angle 2.99°	35°35'04.15"N 129°21'08.95"E
USN VOR/DME		35°35'54.6"N 129°21'11.8"E
SEDMO	BRG 352.68° / 13.00 NM USN	35°48'27.1"N 129°16'55.0"E